Geophysical Imaging

Homework 1: Seismic Methods and Seismic Waves

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1 Definition of the Project

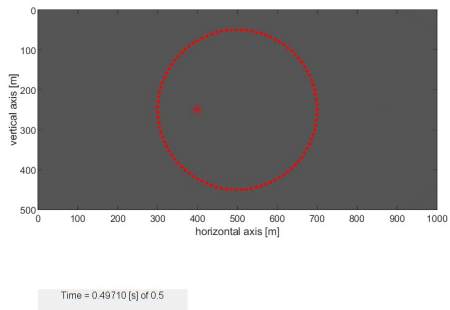
This project investigates two-dimensional acoustic wave propagation using a finite-difference numerical solver. The `ACU2D_PRO` simulator is used to model seismic wave propagation by varying subsurface velocity models, source parameters, and receiver configurations. The resulting pressure fields and synthetic seismograms are analyzed to highlight fundamental wave-propagation phenomena and to assess how acquisition geometry and medium properties control the recorded data.

2 Exercise 1

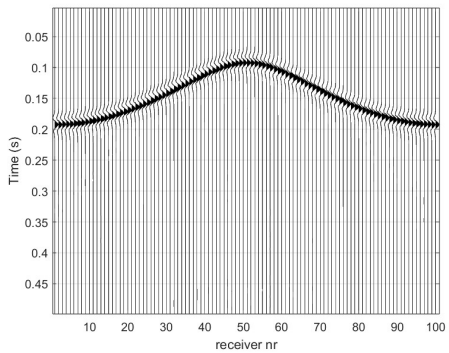
2.1 Reference Case (No Change)

The reference results were obtained by running **Example A1** without modifying the provided code or model parameters. The simulation describes acoustic wave propagation in a homogeneous two-dimensional medium characterized by a constant velocity. As expected for an isotropic medium, the wavefield propagates radially from the source, producing nearly circular wavefronts.

Receivers are distributed along a circular array, and the recorded seismic traces are dominated by the direct-wave arrival. Variations in arrival time along the receiver array are due to different source–receiver distances because the source is not exactly at the circle center. The observed travel times are consistent with constant-velocity propagation.



(a) Wavefield snapshot (homogeneous reference case).



(b) Recorded traces (circular receiver array).

Figure 1: Exercise 1 reference results.

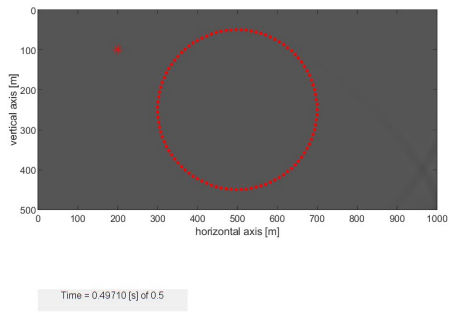
Table 1: Reference case parameters (Exercise 1).

Parameter	Reference case
Source position (x, z) [m]	$(400, 250)$
Receiver geometry	Circular array ($r = 200$ m, center $(500, 250)$ m)
Source type	Ricker
Velocity [m/s]	2000 (constant)

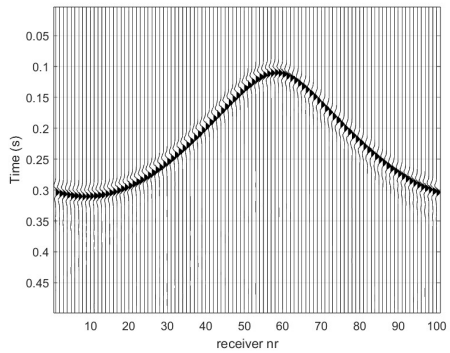
2.2 Changing Source Position

The seismic source is moved from a position close to the receiver-array center to a location outside the circular geometry, while all other parameters remain unchanged. This breaks acquisition symmetry and increases source–receiver distances.

Consequently, recorded traces show later arrival times and a broader travel-time spread than in the reference case. Because the velocity model is unchanged, the differences are purely geometric.



(a) Wavefield snapshot (moved source).



(b) Recorded traces (moved source).

Figure 2: Exercise 1 results after changing the source position.

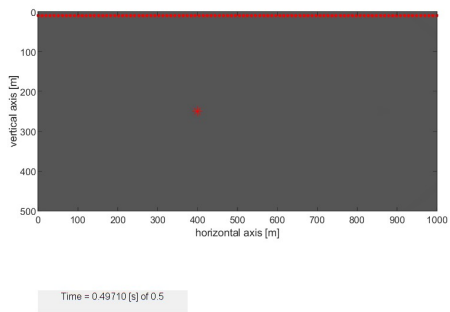
Table 2: Effect of changing the source position (Exercise 1).

Parameter	Reference case	Modified case
Source position (x, z) [m]	$(400, 250)$	$(200, 100)$
Receiver geometry	Circular array	Circular array
Source type	Ricker	Ricker
Velocity [m/s]	2000	2000

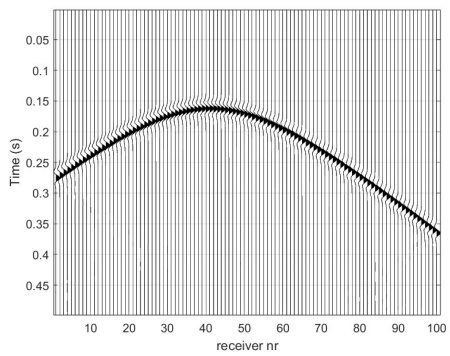
2.3 Changing Receiver Geometry

The receiver geometry is modified from a circular array to a linear receiver line near the surface, while the source and velocity model remain unchanged. This corresponds to a classical shot-gather geometry.

The seismogram displays a direct-arrival moveout approximately symmetric around the receiver closest to the source. This highlights how receiver distribution controls the appearance and interpretability of seismic records.



(a) Wavefield snapshot (line receivers).



(b) Recorded traces (line receivers).

Figure 3: Exercise 1 results after changing the receiver geometry.

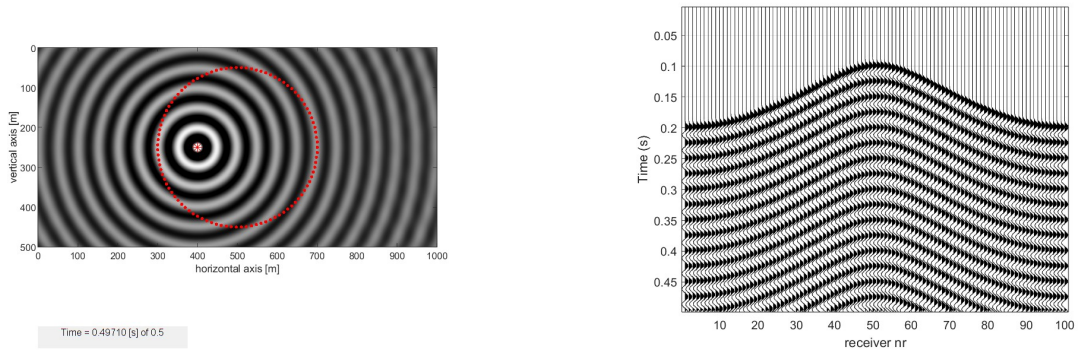
Table 3: Effect of changing receiver geometry (Exercise 1).

Parameter	Reference case	Modified case
Source position (x, z) [m]	$(400, 250)$	$(400, 250)$
Receiver geometry	Circular array	Linear line near surface ($z \approx 20$ m)
Source type	Ricker	Ricker
Velocity [m/s]	2000	2000

2.4 Changing Source Type

The source type is changed from a Ricker wavelet to a sinusoidal source, while all other parameters remain fixed. Unlike the short broadband Ricker pulse, the sinusoidal source emits energy continuously at a single frequency.

The pressure field exhibits multiple repeating wavefronts, and the recorded traces become periodic rather than impulsive.



(a) Wavefield snapshot (sinusoidal source).

(b) Recorded traces (sinusoidal source).

Figure 4: Exercise 1 results after changing the source type.

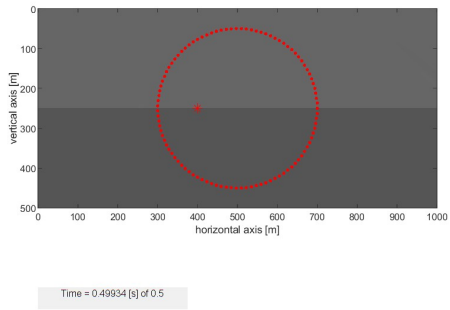
Table 4: Effect of changing the source type (Exercise 1).

Parameter	Reference case	Modified case
Source type	Ricker	Sinusoidal
Receiver geometry	Circular array	Circular array
Velocity [m/s]	2000	2000

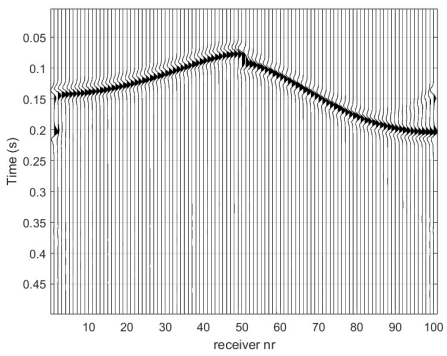
2.5 Two-Layer Velocity Model

The homogeneous velocity model is replaced by a two-layer structure with $V_1 = 1800$ m/s above $V_2 = 3000$ m/s, separated by a horizontal interface at $z = 250$ m. The source and receivers are unchanged.

The wavefield shows reflection and transmission at the interface, and the seismogram includes additional arrivals associated with the velocity contrast.



(a) Wavefield snapshot (two-layer model).



(b) Recorded traces (two-layer model).

Figure 5: Exercise 1 results for a two-layer velocity model.

Table 5: Effect of introducing a two-layer velocity model (Exercise 1).

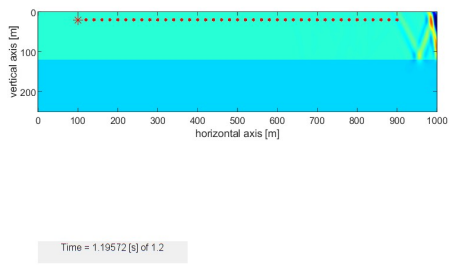
Parameter	Reference case	Two-layer case
Velocity model	2000 m/s (constant)	$V_1 = 1800$ m/s (upper) $V_2 = 3000$ m/s (lower)
Interface depth [m]	–	250

3 Exercise 2

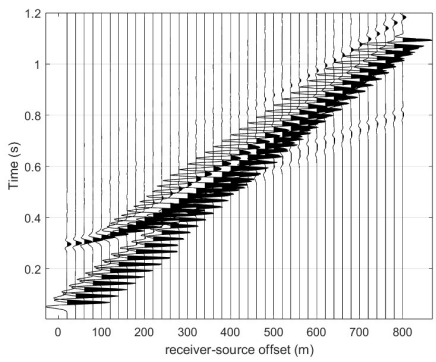
3.1 Reference Configuration

A horizontally layered acoustic medium composed of two constant-velocity layers is simulated using a surface receiver line. The wavefield snapshot highlights wave propagation within each layer as well as interactions at the velocity interface.

The recorded seismogram exhibits distinct arrivals associated with direct waves, reflected waves, and refracted (head) waves. The upper layer is characterized by a constant P-wave velocity V1=800 m/s while the lower layer has a higher constant velocity V2=1500 m/s. This velocity increase with depth produces reflection and refraction phenomena observable in the recorded seismograms. Furthermore, we know that slope has an inverse relationship with velocity here.



(a) Wavefield snapshot



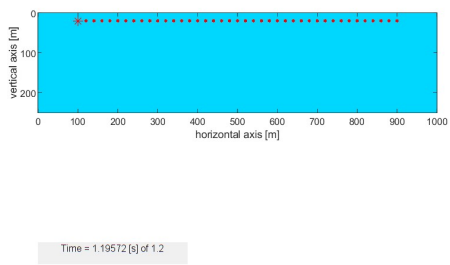
(b) Recorded traces

Figure 6: Exercise 2 reference results .

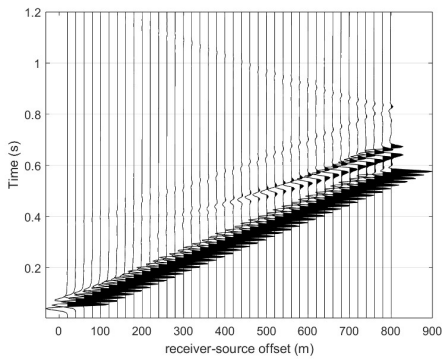
3.2 Modified Configuration

Using a homogeneous acoustic medium with constant velocity, we modify the acquisition geometry to extend the receiver offsets and analyze how recorded signal strength decreases with source–receiver distance. The shot gather still shows a single coherent first-arrival event with approximately linear moveout, as expected in a homogeneous medium. For each receiver trace, we compute the trace energy (e.g., sum of squared amplitudes over time) and plot energy versus offset. The resulting trend primarily reflects geometrical spreading (and, if included in the simulation, intrinsic attenuation), while any weak late arrivals at the largest offsets are attributable to numerical boundary effects rather than subsurface interfaces.

The important point that we should consider here is that we have received several signals for some of our receivers, and this is because of the position of our source, which is located at the surface.



(a) Wavefield snapshot (modified geometry).



(b) Recorded traces (modified geometry).

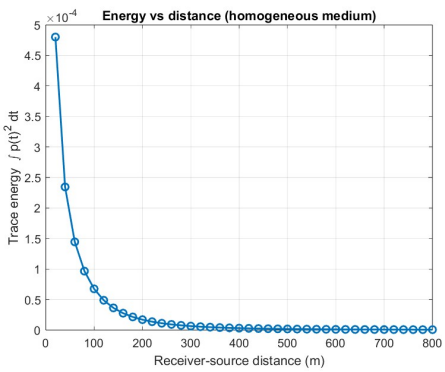
Figure 7: Exercise 2 results after modifying the acquisition geometry.

3.3 Trace Energy Versus Distance

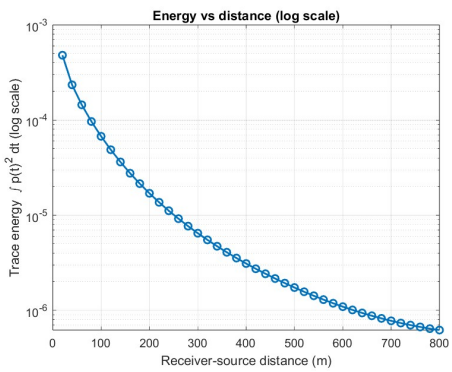
Trace energy is computed as a discrete approximation of

$$E = \sum_t p(t)^2 \, \Delta t,$$

where $p(t)$ is recorded pressure and Δt is the sampling interval.



(a) Energy vs. distance (linear scale).



(b) Energy vs. distance (log scale).

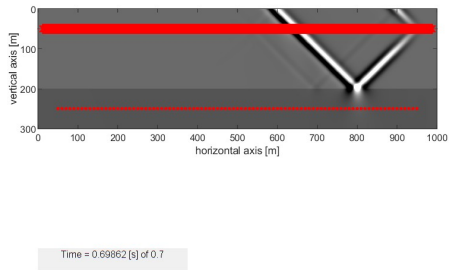
Figure 8: Exercise 2 trace-energy analysis.

Energy decreases with source–receiver distance, consistent with geometric spreading.

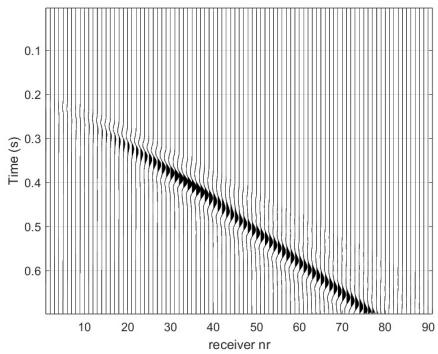
4 Exercise 3

4.1 Reference Example (Plane Wave in a Two-Layer Medium)

A time-delayed shallow source line generates an approximately plane wave propagating downward. At the interface, energy splits into transmitted and reflected components, illustrating refraction and reflection. Sources are located at the upper layer and receivers are located at the bottom layer. When a plane wave encounters an interface between two media with different velocities, reflected and transmitted waves are generated, obeying Snell's law. The wave reaches all receivers with the same incidence angle and constant horizontal slowness.



(a) Wavefield snapshot (plane wave; two-layer model).

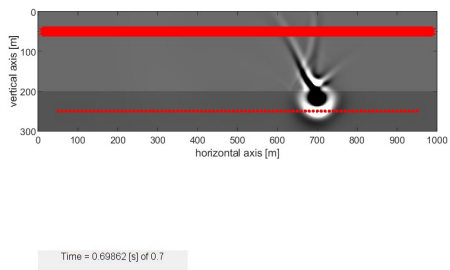


(b) Recorded traces (plane-wave experiment).

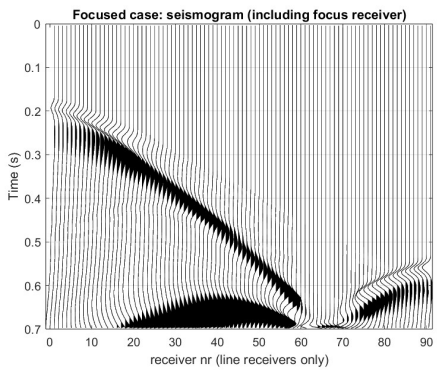
Figure 9: Exercise 3 reference: plane-wave propagation and interface interaction.

4.2 Modified Example (Beamforming and Focusing)

Time delays are applied to focus energy at $(x_f, z_f) = (700 \text{ m}, 250 \text{ m})$. The snapshot shows a strong localized amplitude increase near the focus. The wavefield snapshot demonstrates constructive interference of wavefronts at the prescribed focus location due to the applied source time delays, resulting in a strong localized amplitude increase. The corresponding seismogram shows a maximum response at receivers near the focus, confirming successful beamforming and spatial focusing of acoustic energy.



(a) Focused wavefield snapshot (convergence at the target).

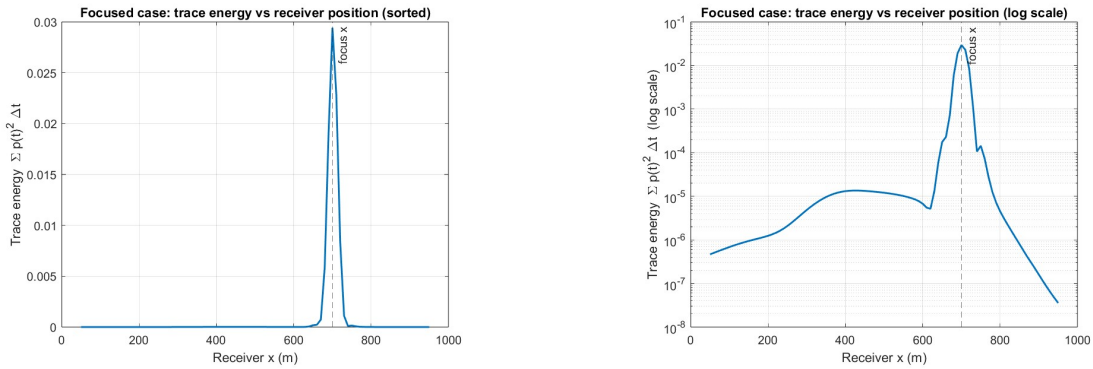


(b) Focused seismogram (maximum response near the focus).

Figure 10: Exercise 3 modified: focusing via source time delays (beamforming).

4.3 Trace Energy Versus Receiver Position (Focused Case)

$$E = \sum_t p(t)^2 \, \Delta t.$$



(a) Trace energy vs. receiver position (linear scale). (b) Trace energy vs. receiver position (log scale).

Figure 11: Exercise 3 focused case: energy distribution along the receiver line.

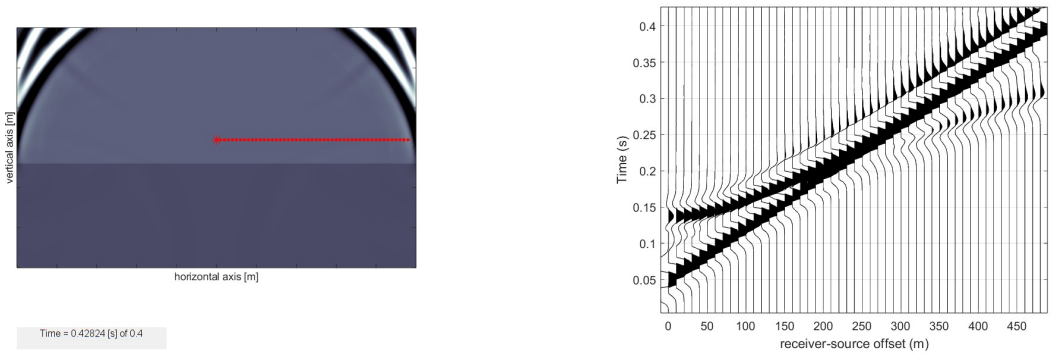
4.4 Comparison and Interpretation

Compared to the reference case, focusing alters the spatial energy distribution by forcing in-phase arrivals at the target, producing constructive interference at the focus and reduced energy elsewhere. This demonstrates that focusing is controlled by acquisition design (source timing/geometry) in addition to medium properties.

5 Exercise 4

5.1 Results of the Seismic Refraction Simulation

The seismic refraction experiment was simulated using the provided `.p` file, producing a wavefield snapshot and a shot gather. The wavefield shows circular wavefronts radiating from the source; interaction with a velocity contrast generates critically refracted (head) waves that return to the surface at larger offsets. Furthermore, we know that wavelength has a direct relationship with velocity (here the frequency is constant). By paying attention to the wavefield snapshot, we can understand that a lower layer has a higher wave length, and so it has higher velocity.



(a) Wavefield snapshot (refraction model). (b) Shot gather (direct and refracted arrivals).

Figure 12: Exercise 4 simulation outputs for the refraction experiment.

The shot gather displays two first-arrival branches:

- **Direct wave** at small offsets, traveling in the near-surface low-velocity layer.
- **Refracted (head) wave** at larger offsets, traveling along a deeper high-velocity layer.

5.2 Manual Picking of First Arrivals

First-arrival times were manually picked from the shot gather. Direct-wave arrivals were picked at near offsets. Refracted-wave arrivals were picked at larger offsets, where the head wave arrives earlier than the direct wave.

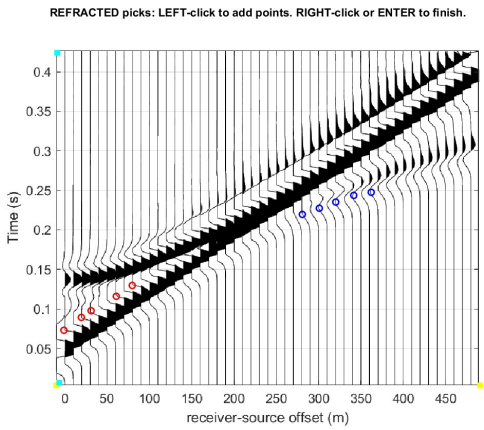


Figure 13: Shot gather with manually picked first arrivals (direct and refracted branches).

5.3 Calibration of the Shot Gather Image

Because the shot gather was available as an image rather than raw numerical traces, pixel coordinates were calibrated to physical units. Four reference points were selected: $x = 0$ m, $x = 450$ m, $t = 0$ s, and $t = 0.4$ s. Linear mappings were used:

$$x_{\text{phys}} = a_x \, x_{\text{pix}} + b_x, \tag{1}$$

$$t_{\text{phys}} = a_t \, y_{\text{pix}} + b_t. \tag{2}$$

5.4 Time–Distance (t–x) Analysis

Calibrated picks were plotted in a t–x diagram and fitted with straight lines:

$$t = a + bx, \tag{3}$$

where $b = dt/dx$ and $v = 1/|b|$.

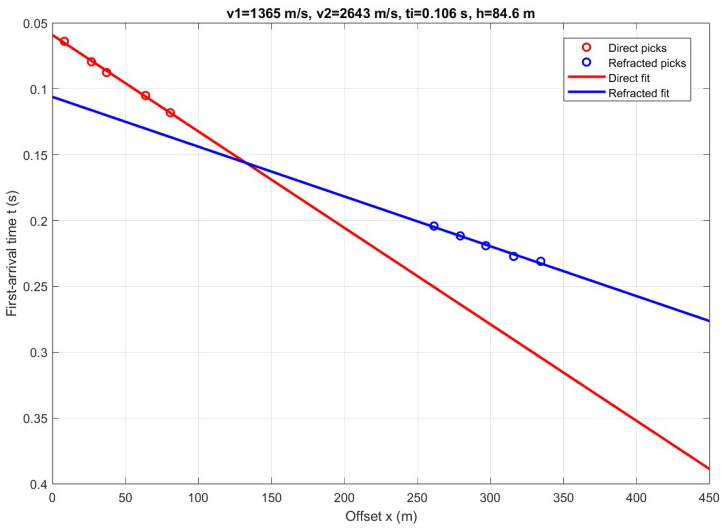


Figure 14: t–x plot of picked first arrivals with linear fits for direct and refracted branches.

Estimated velocities:

$$v_1 = \frac{1}{|b_1|} = 1365 \text{ m/s}, \tag{4}$$

$$v_2 = \frac{1}{|b_2|} = 2643 \text{ m/s}. \tag{5}$$

5.5 Time-Intercept Method and Depth Estimation

The refracted-line intercept is

$$t_i = 0.106 \text{ s}.$$

Assuming a horizontal two-layer model, the refractor depth is

$$h = \frac{t_i v_1 v_2}{2\sqrt{v_2^2 - v_1^2}} = 84.6 \text{ m}.$$

5.6 Discussion and Interpretation

The condition $v_2 > v_1$ supports the existence of a refracting interface. The separation between direct and refracted arrivals is consistent with refraction theory, and the estimated velocities and depth are plausible for shallow investigations.

Table 6: Exercise 4 results from the time-intercept method.

Parameter	Value
Direct-wave velocity v_1	1365 m/s
Refracted-wave velocity v_2	2643m/s
Time intercept t_i	0.106 s
Depth to refractor h	84.6 m

6 Exercise 5

6.1 Huygens' Principle: Wave Propagation Around a High-Velocity Obstacle

Model description Acoustic wave propagation was simulated in a two-dimensional medium using a finite-difference solution of the acoustic wave equation. The background medium has a constant P-wave velocity of 600 m/s. A rectangular high-velocity body (3500 m/s) is embedded between depths of 80–120 m and horizontal distances of 200–300 m. Absorbing boundaries were applied to limit edge reflections.

A point source emitting a Ricker wavelet with central frequency 30 Hz was placed at $(x, z) = (200, 20)$ m. Receivers were positioned along the bottom of the model ($z = 200$ m), spaced every 10 m.

Wavefield snapshot Figure 15 shows the acoustic pressure field. The incident wavefront propagates radially in the background medium. At the obstacle, energy is transmitted and diffracted, and curved wavefronts appear around the block corners, consistent with Huygens' principle (secondary wavelets).

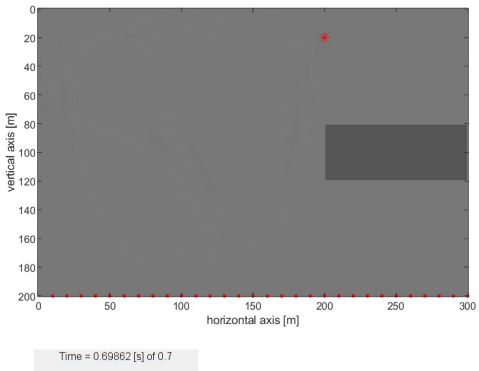


Figure 15: Wavefield snapshot showing diffraction around a high-velocity rectangular obstacle.

Seismic traces Figure 16 shows the recorded traces at $z = 200$ m. First arrivals follow a coherent moveout pattern dominated by the background velocity; scattered energy and travel-time perturbations are vis

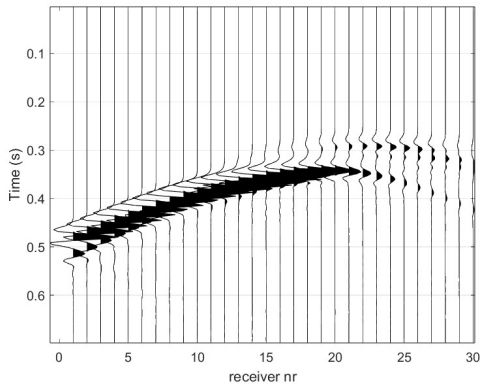


Figure 16: Recorded traces illustrating direct arrivals and scattered energy caused by the high-velocity obstacle.

Interpretation This example demonstrates diffraction and interference in the presence of strong lateral velocity contrasts, showing why wave-based modelling is essential when straight-ray

assumptions fail.

6.2 Image-Based Velocity Model and Acoustic Wave Simulation

Velocity model construction A grayscale image was converted to a velocity model by mapping normalized intensity $I(x, z) \in [0, 1]$ to velocity:

$$v(x, z) = v_{\min} + I(x, z)\,(v_{\max} - v_{\min}),$$

with $v_{\min} = 1500$ m/s and $v_{\max} = 4500$ m/s, over a physical domain of 7.5 km (x) by 2.5 km (z).

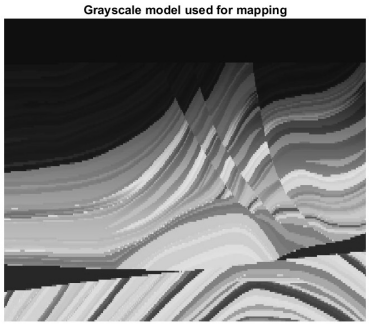
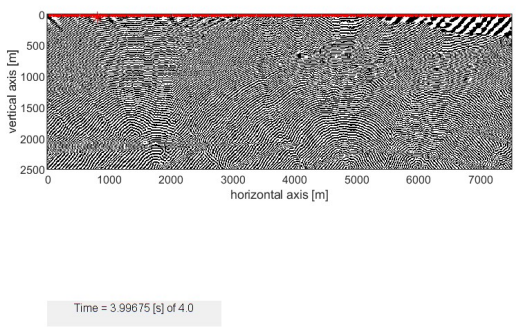


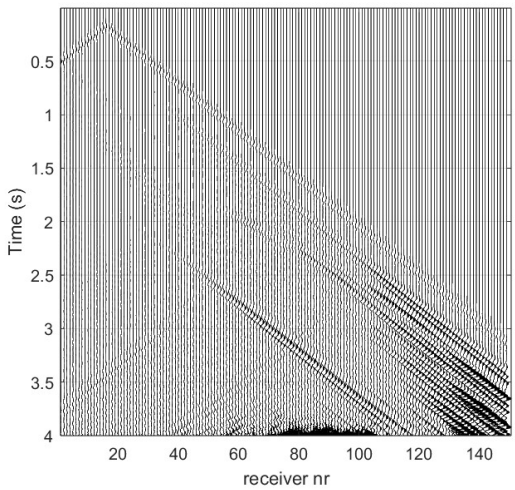
Figure 17: Grayscale input image used to build the velocity model (brighter = higher velocity).

Source and acquisition geometry A Ricker source with $f_0 = 10$ Hz was placed at $(x, z) = (800, 50)$ m. Receivers were located near the surface at $z = 20$ m with 50 m spacing.

Wavefield behavior and recorded data Propagation through the heterogeneous model distorts wavefronts and generates scattered energy.



(a) Input image mapped to velocity (brighter = higher velocity).



(b) Wavefield snapshot in the image-based heterogeneous velocity model (distorted wavefronts and scattering).

Figure 18: Image-based velocity model and simulated wavefield.

7 Exercise 6

In this exercise, a smartphone equipped with inertial sensors was used to measure linear acceleration during walking and traveling by bus. Data were acquired with the *Phyphox* application, which records a_x, a_y, a_z and the absolute acceleration magnitude.

7.1 Walking

Acceleration signals show rapid fluctuations consistent with the cyclic nature of gait (the way a person walks or moves their body in a repeating pattern). Approximate peak values:

- $a_x \approx -0.77$ m/s^2,
- $a_y \approx 1.46$ m/s^2,
- $a_z \approx 1.48$ m/s^2,

and

$$a_{\text{abs}} \approx 2.21 \text{ m/s}^2.$$

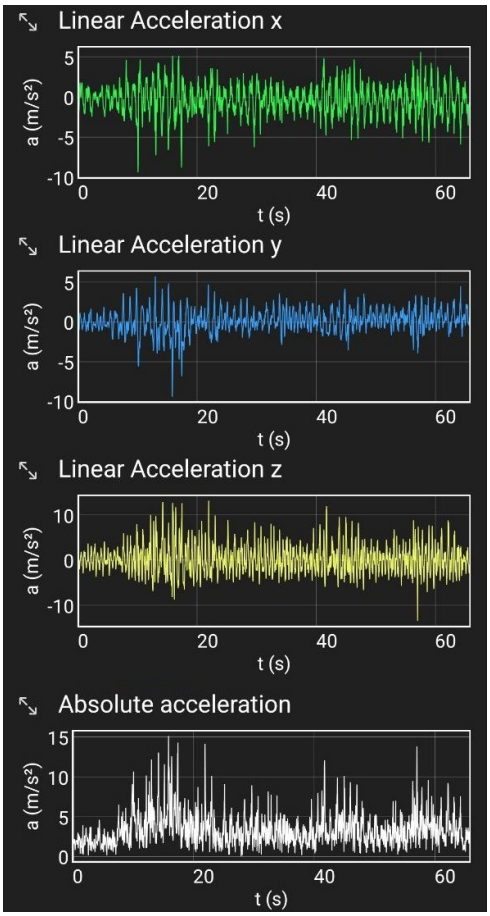


Figure 19: Exercise 6 walking measurements (absolute acceleration).

7.2 Bus

Bus motion produces smoother, lower-frequency variations dominated by acceleration/braking and road vibrations. Approximate peak values:

- $a_x \approx 0.38$ m/s^2,

- $a_y \approx 0.02$ m/s^2,
- $a_z \approx -0.33$ m/s^2,

and

$$a_{\text{abs}} \approx 0.50 \text{ m/s}^2.$$

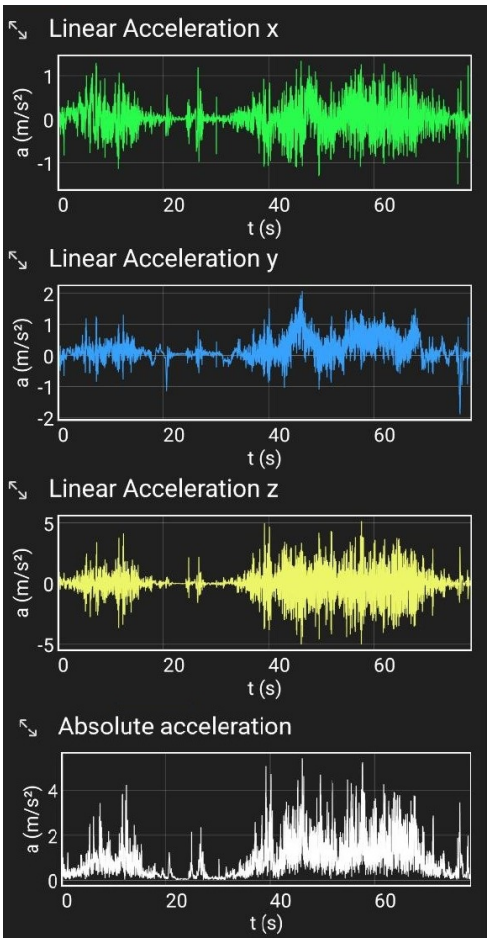


Figure 20: Exercise 6 Bus measurements (acceleration component).

7.3 Comparison and Discussion

Walking produces larger, more irregular accelerations due to active body motion, while bus motion produces smaller, smoother accelerations. The absolute acceleration during walking is more than four times larger than during the bus ride, demonstrating the sensitivity of smartphone sensors for basic motion classification.

7.4 Conclusion

Smartphone accelerometers provide a practical tool for collecting real-world motion measurements. Walking is characterized by high-amplitude, rapidly varying accelerations, whereas bus travel shows lower-amplitude and smoother signals dominated by vehicle dynamics.